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Construction of 3-Azabicyclo[3.1.0]hexane Backbone by the Reaction of Allenes with Allylamines *via* Tandem Michael Addition and Copper-Mediated Oxidative Carbanion Cyclization

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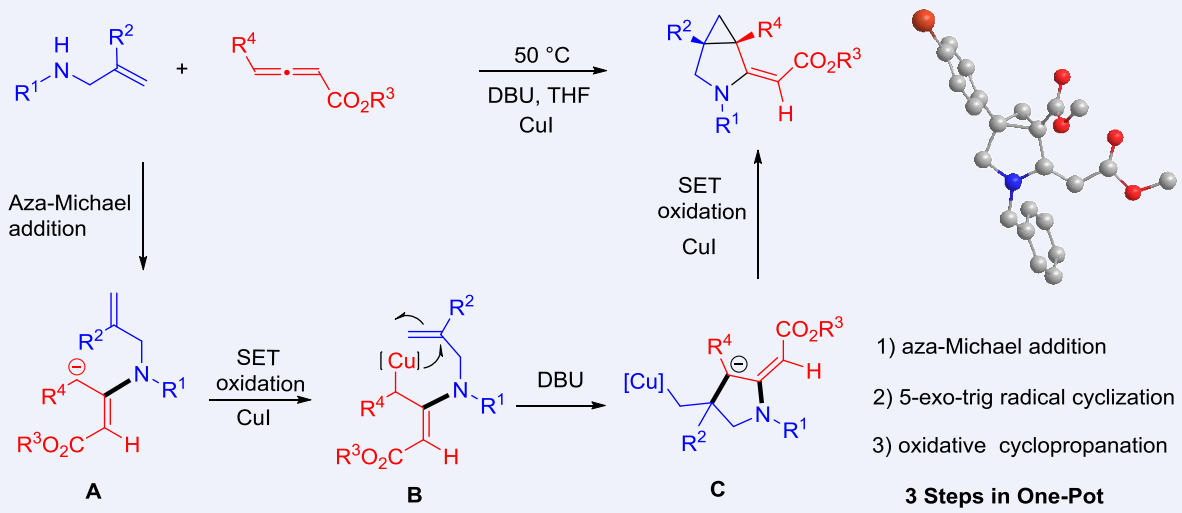
Keywords

Oxidative coupling | Allenes | Allylamine | 3-Azabicyclo[3,1,0]hexane | Miachel addition

Main observation and conclusion

A facile synthetic method for the construction of 3-azabicyclo[3.1.0]hexane in the presence of copper catalyst system was developed. The reaction proceeds through Michael addition of allylamines with allenes followed by copper-mediated intramolecular oxidative carbanion 5-exo-trig radical cyclization, affording potential biologically active 3-azabicyclo[3.1.0]hexane derivatives in moderate to high yields (42%—85%).

Comprehensive Graphic Content



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Background and Originality Content

3-Azabicyclo[3.1.0]hexane backbone is widely found as key structural unit in various natural products, pharmaceutically active molecules and drugs (Figure 1),[1] such as Bicifadine, a drug candidate for the treatment of low back pain in the Phase III clinical phase;[2] selective dopamine D3 receptor (DRD3) antagonist (GSK598809) for cocaine use disorder;[3] Boceprevir for the treatment of type I chronic hepatitis disease, ketone hexanone kinase (KHK) inhibitor for the treatment of non-alcoholic fatty liver dis-ease (NAFLD);[4] (+)-duocarmycin A, a kind of effective antibiotic with anti-tumor activity;[5] Procymidone (procymidone), a new type of fungicide.[6]

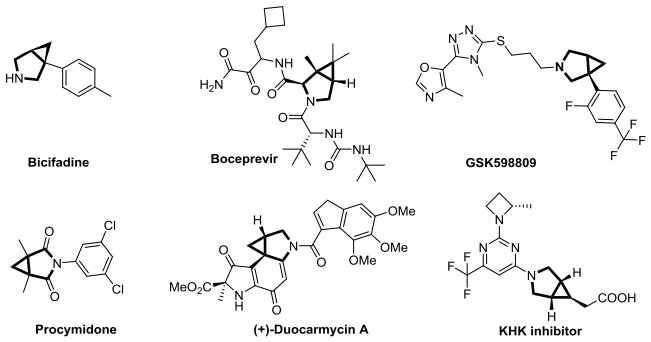


Figure 1 Bioactive molecules with 3-azabicyclo[3.1.0]hexane scaffold.

Owing to their wide-ranging applications in organic synthesis and drug discovery, several synthetic methods of 3-azabicyclo-[3.1.0]hexane have been reported.[7] Meanwhile, several examples of gold-catalyzed oxidative cyclopropanation have also been reported.[8]

On the other hand, allene derivatives are a class of highly important versatile building blocks in organic synthesis, which can be employed as special Michael acceptors for the reactions with different nucleophilic heteroatoms to construct corresponding heterocycles.[9-10] Meanwhile, we recently reported a series of direct α,β-double electrophilic reactions by the treatment of allene-1,3-dicarboxylic esters with 1,3-dicarbonyl derivatives or enamines under a mild I^-/TBHP or H_2O_2 reaction system to afford the multi-substituted furans, thiophens and pyrroles, respectively.[11-12] Recently, Baskaran[13] *et al*. reported an efficient multi-component reaction for one-pot synthesis of cyclic amidine derivatives with biologically important 3-azabicyclo[3.1.0]alkane scaffold by the reaction of ketenimine **I** and allylamines (Scheme 1a). Therefore, as part of our continuing efforts to develop synthetic methods for the formation of heterocycles,[14] we envisaged that the use of allene as an alternative substrate instead of ketenimine **I** (Scheme 1a) would provide a new synthetic method for accessing 3-azabicyclo[3.1.0]hexane scaffold. Herein, we would like to describe a novel synthetic access to 3-azabicyclo[3.1.0]hexane derivatives by the reaction of allylamines with allenes under 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), affording carbanion **A** similar to carbanion **II** in Baskaran's work, *via* tandem Michael addition and copper-mediated oxidative carbanion cyclization (Scheme 1b).

Results and Discussion

Diallylamine **1a** and allene **2a** as model substrates were selected to verify the feasibility of this reaction (Table 1). The reaction occurred in the presence of two equivalents of $CuBr_2$ to give the expected product **3aa**, but it was accompanied with byproduct **4aa** with yields of 17% and 45%, respectively (Table 1, entry 1). Fortunately, when the loading of $CuBr_2$ was reduced to 1.5 equiv, the yield of byproduct **4aa** decreased significantly, with increased yield of **3aa** (Table 1, entry 2), which suggested that the excess of $CuBr_2$ will further lead to the bromination of **3aa**, thus a series of bromine-free catalysts were next screened (Table 1, entries 3—7). $FeCl_3$ and $Cu(OAc)_2$ failed to deliver product **3aa**, and CuI (2.0 equiv) was found to be an optimal catalyst to give corresponding product **3aa** in 80% yield (Table 1, entry 7).

Screening of solvents (Table 1, entries 7—13) revealed that THF was the optimal reaction medium to afford the corresponding product **3aa** in 80% yield. Subsequently, different bases were investigated (Table 1, entries 14—18), 2.2 equiv of DBU was found to give optimal yield. Decreasing the temperature to 30 °C gives comparable yield of product **3aa** (Table 1, entry 19).

Having identified the optimal reaction conditions, the reaction was extended to a variety of different substituted allylamines **1**. Unfortunately, substrate allylamines with 2-aryl group gave the

Scheme 1 Strategies for synthesis of structurally diverse heterocycles

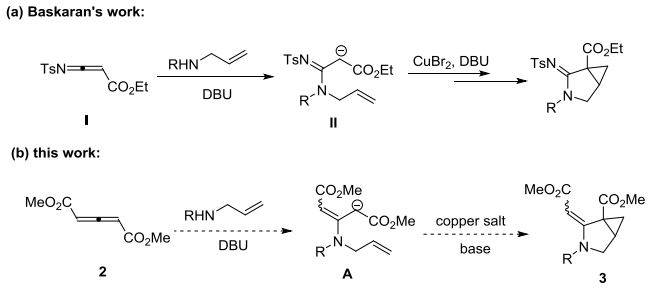


Table 1 Optimization of reaction conditions



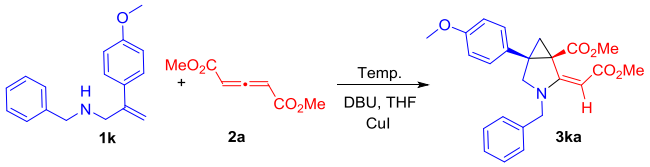
Entry[a]	Catalyst/ equiv	Solvent	Base/equiv	Temp./ °C	Yield[b]/% **3aa**	Yield[b]/% **4aa**
1	$CuBr_2$ (2)	THF	DBU (2.2)	60	17	45
2	$CuBr_2$ (1.5)	THF	DBU (2.2)	60	50	14
3	CuI (1.5)	THF	DBU (2.2)	60	72	—
4	$FeCl_3$ (1.5)	THF	DBU (2.2)	60	—	—
5	$Cu(OAc)_2$ (1.5)	THF	DBU (2.2)	60	trace	—
6	CuI (2.5)	THF	DBU (2.2.)	60	68	—
7	CuI (2)	THF	DBU (2.2)	60	80	—
8	CuI (2)	1,4-dioxane	DBU (2.2)	60	43	—
9[c]	CuI (2)	CH_2Cl_2	DBU (2.2)	reflux	40	—
10	CuI (2)	DCE	DBU (2.2)	60	57	—
11[c]	CuI (2)	diethyl ether	DBU (2.2)	reflux	56	—
12	CuI (2)	toluene	DBU (2.2)	60	45	—
13	CuI (2)	CH_3CN	DBU (2.2)	60	52	—
14	CuI (2)	THF	DMAP (2.2)	60	56	—
15	CuI (2)	THF	DABCO (2.2)	60	—	—
16	CuI (2)	THF	dibutylamin (2.2)	60	55	—
17	CuI (2)	THF	NaOH (2.2)	60	—	—
18	CuI (2)	THF	Cs_2CO_3 (2.2)	60	—	—
19[d]	CuI (2)	THF	DBU (2.2)	30	78	—

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.2 mmol), solvent (1.5 mL), base (0.44 mmol), 60 °C, under air atmosphere. [b] Isolated yield; [c] Reaction was conducted at reflux temperature. [d] Reaction was conducted at 30 °C.

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product in lower yields, so we chose **1k** and **2a** as model substrates for further optimization of reaction conditions (Table 2). First, the ratio of **1k** and **2a** was investigated and 1 : 1.1 was found to be an optimal ratio (Table 2, entries 1—5). Subsequently, the loading of DBU (Table 2, entries 6—9) and CuI was further screened (Table 2, entries 10—13), which determined that 2.8 equiv of DBU and 2.4 equiv of CuI were the optimal conditions. Finally, the reaction temperature was investigated (Table 2, entries 14—17). 50 °C was found to give the optimal yield (54% yield). Thus, the combination of **1k** (1.0 equiv) and **2a** (1.1 equiv) in the presence of 2.8 equiv DBU and 2.4 equiv CuI in THF at 50 °C was determined as the optimal reaction conditions, providing **3ka** in 54% yield (Table 2, entry 15).

Table 2 Further screening of reaction conditions



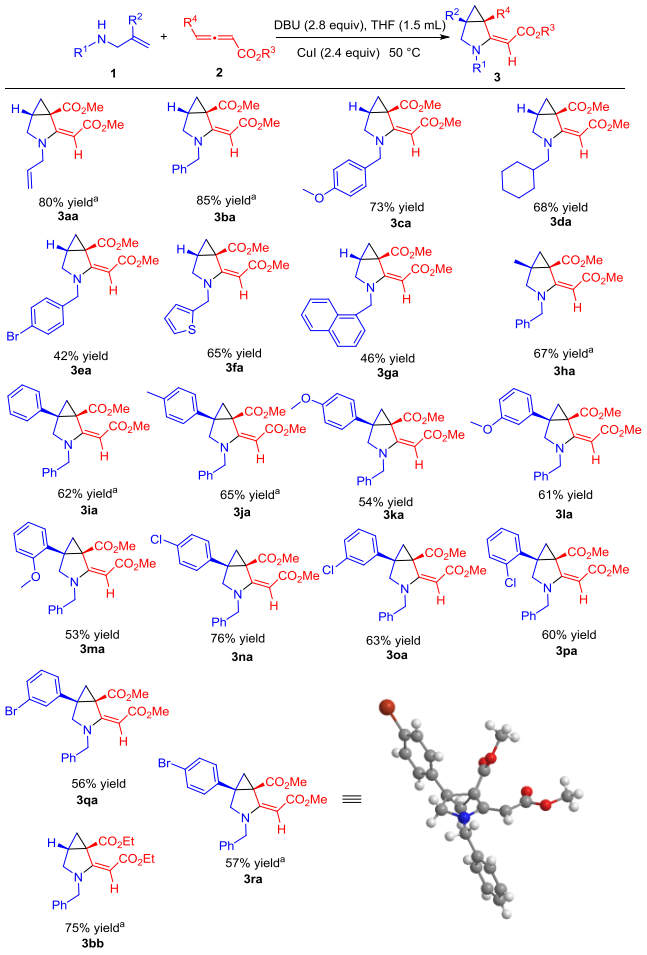
Entry[a]	**1k** : **2a** (ratio)	DBU/equiv	CuI/equiv	Temp./°C	Yield[b]/%
1	1 : 1	2.2	2	30	34
2	1 : 1.1	2.2	2	30	40
3	1 : 1.3	2.2	2	30	40
4	1 : 1.5	2.2	2	30	40
5	1 : 2	2.2	2	30	40
6	1 : 1.1	2.5	2	30	42
7	1 : 1.1	2.8	2	30	48
8	1 : 1.1	3	2	30	48
9	1 : 1.1	3.5	2	30	47
10	1 : 1.1	2.8	2.2	30	49
11	1 : 1.1	2.8	2.4	30	51
12	1 : 1.1	2.8	2.7	30	45
13	1 : 1.1	2.8	3	30	42
14	1 : 1.1	2.8	2.4	40	50
15	1 : 1.1	2.8	2.4	50	54
16	1 : 1.1	2.8	2.4	70	48
17	1 : 1.1	2.8	2.4	90	43

[a] Reaction conditions: **1k** (0.2 mmol), THF (1.5 mL), under air atmosphere. [b] Isolated yield.

Having established the optimal reaction conditions, the substrate scope of this process was investigated (Scheme 2). The reaction of allylamines **1a**—**1g** with different substituents at the N-position underwent smoothly, affording corresponding products **3aa**—**3ga** in moderate to good yields (42%—85%). Encouraged by these results, various substituents R^2 of allylamines **1h**—**1i** were then investigated, and it was found that allylamine with either alkyl or aryl group gave corresponding products **3ha**—**3ra** in good yields (62%—67%). Further extension of substrate scope of allylamine with electron-withdrawing or electron-donating groups of aromatic substituents **1j**—**1r** all reacted smoothly to furnish the products **3ja**—**3ra** in good to high yields (53%—76%). Finally, allene **2b** with ethyl ester was also found well tolerated, affording **3bb** in 75% yield. In addition, the structure of **3ra** was established by single crystal X-ray diffraction analysis (see Supporting Information).

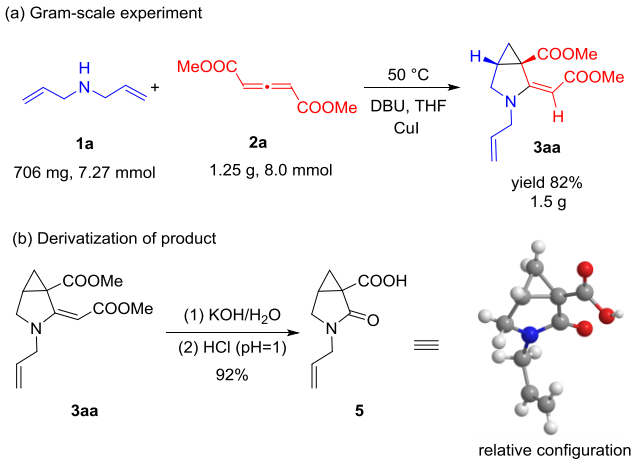
To demonstrate the utility of this strategy, a gram scale reaction was carried between allylamine **1a** and allene **2a**, providing **3aa** in 82% yield (Scheme 3a). On the other hand, an unexpected carboxylic acid derivative **5** of 3-azabicyclo[3.1.0] hexane can be easily obtained in 92% yield by the treatment of **3aa** with alkaline

Scheme 2 Investigation of substrate scope of allylamine and allene



[a] Reaction conditions: **1** (0.2 mmol), **2** (0.2 mmol), THF (1.5 mL), DBU (0.44 mmol), CuI (0.4 mmol), 30 °C, under air atmosphere.

Scheme 3 Demonstration of synthetic utility



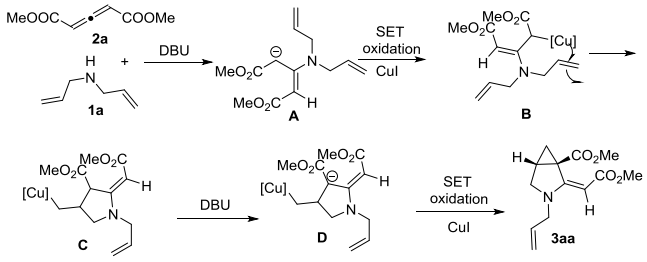
aqueous solution (Scheme 3b). In addition, the structure of **5** was established by single crystal X-ray diffraction analysis (see Supporting Information).

A plausible mechanism for the formation of the 3-azabicyclo-[3.1.0]hexane **3aa** is shown in Scheme 4, which is proposed according to a similar mechanism reported by Baskaran.[8] The reaction of allylamine **1a** and allene **2a** generates a carbanion **A** *via* aza-Michael addition, which is then subjected to a SET oxidation

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with CuI, followed by a 5-exo-trig radical cyclization to afford intermediate **C**. Further deprotonation in the presence of DBU and subsequent additional SET oxidation lead to a cyclopropanation to produce 3-azabicyclo[3.1.0]hexane **3aa**.

Scheme 4 Proposed mechanism



Conclusions

In summary, we have demonstrated a facile synthetic method for the construction of 3-azabicyclo[3.1.0]hexane scaffold by the reaction of allylamine with electron-deficient allene through tandem aza-Michael addition and intramolecular oxidative carbanion 5-exo-trig radical cyclization and cyclopropanation in the presence of CuI. By using this novel synthetic method, a series of potential biologically active 3-azabicyclo[3.1.0] hexane derivatives **3aa—3bb** were obtained in moderate to high yields (42%—85%). A gram-scale synthesis of **3aa** and its further transformation under alkaline aqueous solution led to an unexpected carboxylic acid derivative **5** with an 3-azabicyclo[3.1.0]hexane skeleton, which suggests potential application in a medicinal chemistry scenario.

Experimental

Allylamine **1** (0.2 mmol) and allene **2** (0.22 mmol) were added to THF (1.5 mL) at 50 °C. The mixture was then stirred at the same temperature until the reaction was nearly completed monitored by the TLC. Afterwards, CuI (0.48 mmol) and DBU (0.56 mmol) were added to the mixture at 50 °C. After the completion of the reaction indicated by TLC, the reaction mixture was concentrated under reduced pressure. The resulted residue was purified through flash chromatography on silica gel (petroleum ether/ethyl acetate = 2/1) to afford pure **3**.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202000405.

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